Proof Methods

1. Give a *direct proof* of the following statements.

(a) Recall that, for natural numbers n and k with $k \leq n$, the binomial coefficient is defined as $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$. Show that, if $n \geq 2$ is an integer, that $n^2 = 2\binom{n}{2} + \binom{n}{1}$.

(b) Let a and b be integers. If $a|b$, then $a|\,(3b^3 - 4b^2 + 18b + 2a)$.

(c) If x is an integer and $x^2|x$, then $x \in \{-1, 0, 1\}$.

2. Give a *contrapositive proof* of the following statements.

(a) Suppose x and y are integers. If $3 \nmid xy$, then $3 \nmid x$ and $3 \nmid y$.

(b) (Hammack Problem 5.6) Suppose x is a real number. If $x^3 - x > 0$, then $x > -1$.

(c) (Hammack Problem 5.7) Suppose a and b are integers. If both ab and $a + b$ are even, then both a and b are even.

Prove the following using *proof by contradiction*.

3. $\sqrt{2}$ is irrational.

4. There are infinitely many prime numbers.

5. (Basically Hammack Problem 6.16)[1] Suppose x and y are non-negative real numbers. Then

$$\frac{x+y}{2} \geq \sqrt{xy}$$

or, equivalently, $x + y \geq 2\sqrt{xy}$.

[1]This is also called the AM-GM Inequality, as it says that the Arithmetic Mean $\frac{x+y}{2}$ is always greater than or equal to the Geometeric Mean $\sqrt{xy}$.